BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, RAWALPINDI

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

BIOLOGY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Biology (Objective Type) — Inter (Part-II)-A-2022 (For all sessions) — Paper Code: 8467

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8467

Note: NOTE: Write answers to the questions on the objective answer sheet provided. Four possible answers A,B,C and D to each question are given. Which answer you consider correct, fill the corresponding question with Marker or pen ink on the answer sheet provided.

Q1. The vernolic acid and ricinolic acid can be used as hardness in: [numbered "1.1." as printed]

(A) paints
(B) plastics
(C) Paints and plastics
(D) Rubber

Q2. Darwin's "Origin of species" was published in:

(A) 1840
(B) 1859
(C) 1865
(D) 1890

Q3. Study of relationship of different communities to environment is called:

(A) Synecology
(B) Autecology
(C) Embryology
(D) Zoology

Q4. Which one is not a desert?

(A) Thal
(B) Thar
(C) Sahara
(D) Taiga

Q5. The steady internal state of homeostasis is known as:

(A) Disorder
(B) Disease
(C) Normal health
(D) Abnormal health

Q6. Pressure filtration is associated with the:

(A) Glomerular Part
(B) Collecting tubule
(C) Distal convoluted tubule
(D) Collecting duct

Q7. The diameter of thick filament in muscle is:

(A) 7-8 nm
(B) 4 nm
(C) 16 nm
(D) 10 nm

Q8. At the place of attachment of leaf with the shoot a swollen part is called:

(A) Pitch
(B) Pit
(C) Cortex
(D) Pulvinus

Q9. Intelligence is under the control of:

(A) Cerebrum
(B) Cerebellum
(C) Thalamus
(D) Hypothalamus

Q10. Diploid parthenogenesis may occur in:

(A) Bees
(B) Aphid
(C) Wasp
(D) Honey Bee

Q11. The hypoblast is mainly presumptive:

(A) Endoderm
(B) Epiderm
(C) Mesoder
(D) Blastoderm

Q12. It is one of the prominent structure in the chick embryo of 18 hours:

(A) Primitive streak
(B) Neurocoel
(C) Notochord
(D) Coelom

Q13. Initiation codes for every protein coding gene is AUG which encodes for:

(A) Leucine
(B) Serine
(C) Alanine
(D) Methionine

Q14. The term Bivalent means:

(A) One chromosome
(B) Two chromosomes
(C) Two chromatids
(D) Four Chromosomes

Q15. Mongolism is phenotypically:

(A) Male
(B) Female
(C) Male or Female
(D) None

Q16. How many possible alleles of ABO blood group are present in an individual?

(A) 1 (B) 2

(C) 3 (D) 300

Q17. Dopamine producing cells can be grafted in the brain in order to curve [sic]

(A) Haemophilia (B) Epilepsy

(C) Parkinson's disease (D) Alzheimer's

Biology (Essay Type) — Section-I — Inter (Part-II)-A-2022 (For all sessions)

Time Allowed: 2:40 Hours | **Maximum Marks:** 68

Q2. Write short answers of any eight parts from the following. 2x8=16

(i) Why do marine fishes retain trimethylamine oxide in their body?

(ii) Give the significance of tubular secretion in filterate. [sic]

(iii) How do aldosterone and ADH help in concentration of urine?

(iv) Differentiate between heart wood and sap wood.

(v) What is Nutation?

(vi) Name the bones of pectoral girdle.

(vii) Give the role of foetus during the timing of delivery.

(viii) What is the cause and symptoms of Syphilis?

(ix) Differentiate between climate and weather.

(x) What is Taiga? Give conditions in Taiga.

(xi) Define soil. Give its composition.

(xii) What are the effects of global warming?

Q3. Write short answers of any eight parts from the following. 2x8=16

(i) Discuss the effect of age and emotions on epilepsy.

(ii) Why the Hypothalamus is a main Coordination center?

(iii) What type of behaviour is Kinesis?

(iv) What is Product Rule?

(v) Discuss Dominance relation. Give one example.

(vi) What is Rh-factor?

(vii) What is Gene Pharming? [sic]

(viii) What is cystic fibrosis?

(ix) Discuss importance of Tissue-culture.

(x) What are decomposers?

(xi) What is Food Chain?

(xii) What type of trophic level exists in Food Chain?

Q4. Write short answers of any six parts from the following. 2x6=16 [sic — prints 16]

(i) How increase in length of plant body occurs? Discuss it.

(ii) Can aging be slowed down in human? Comment on it.

(iii) Differentiate between Template and coding strands of DNA during transcription.

(iv) Give role of promoter in transcription.

(v) Why DNA replication cannot complete without DNA Helicase

(vi) Distinguish Convergent Evolution from Divergent Evolution.

(vii) Differentiate between Endangered species and Extinct species. Give examples.

(viii) Differentiate between Karyokinesis and cytokinesis.

(ix) What is crossing over? Give its significance.

Q. Section-II

NOTE: Answer any three questions from the following. 8x3=24

5. (a) How osmoregulation takes place in terrestrial environment? 4

5. (b) Write a note on synecology and autecology. 4

6. (a) Demonstrate the antagonistic working of hinge joint of elbow. 4

6. (b) Describe the process of transcription emphasizing initiation, elongation and termination steps. (Post transcriptional modifications are not required) 4

7. (a) Describe the mechanism of synaptic transmission. 4

7. (b) How acid rain is produced? What are the causes and effects of acid rain? 4

8. (a) Describe human menstrual cycle's primary steps. 4

8. (b) Write a note on Sex determination in plants 4

9. (a) Define regeneration with two examples. Discuss the mechanism of regeneration in planaria and salamander. 4

9. (b) Do the anatomical similarities between species bring any evidence in the support of evolution? If yes, explain. 4